

Module 5 Superconductivity Extension: Carrier Transport, Quasiparticle Poisoning, Offset Charge, and Josephson-Device Response

How the preexisting drift-diffusion carrier-transport module changes in the superconducting region

Physics note for RadiationEffects_proj2

Purpose. Module 5 originally converted electrostatic, defect, and thermal fields into semiconductor carrier transport: mobile electrons and holes obey drift-diffusion equations with generation, recombination, mobility degradation, and current output. This superconductivity extension explains what changes when the radiation-effects problem is moved to cryogenic superconducting-device operation. The main change is that the relevant mobile charge carriers are no longer only ordinary conduction-band electrons and valence-band holes. The device may contain frozen-out semiconductor substrate carriers, trapped charges in dielectrics, superconducting condensate current, quasiparticle diffusion in superconducting films, quasiparticle tunneling across Josephson junctions, and radiation-induced offset-charge motion. This note derives these pieces in a textbook style and translates them into Module 5 modeling requirements.

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Symbols and acronyms used in this document

Symbol / acronym	Name	Meaning in this document
Module 5	Carrier-transport module	Existing module that evolves mobile charge carriers and computes electrical response.
SC	Superconducting	Condensed state below a critical temperature with zero DC resistance in ideal equilibrium and an excitation gap.

Symbol / acronym	Name	Meaning in this document
QP	Quasiparticle	Broken-Cooper-pair excitation in a superconductor.
DD	Drift-diffusion	Semiconductor transport model for mobile electrons and holes.
SRH	Shockley-Read-Hall	Trap-assisted recombination through localized defect states.
JJ	Josephson junction	Weak link between superconductors; supports phase-dependent supercurrent and quasiparticle tunneling.
TLS	Two-level system	Defect-like microscopic degree of freedom that can fluctuate and couple to electric fields or qubits.
CP	Cooper pair	Bound pair of electrons forming the superconducting condensate.
PBP	Pair-breaking phonon	Phonon with energy sufficient to break a Cooper pair, approximately $\hbar\omega \geq 2\Delta$.
$\mathbf{x}$	Position	Spatial coordinate in the 2D or 3D device.
t	Time	Time coordinate.
q, e	Elementary charge magnitude	Positive scalar charge magnitude. In superconductivity notation, the Cooper-pair charge magnitude is $2e$.
ϕ	Electrostatic potential	Scalar potential from Module 2.
$\mathbf{E}$	Electric field	$\mathbf{E} = -\nabla\phi$.
n, p	Semiconductor carriers	Mobile electron and hole concentrations in the baseline Module 5 model.
G_{eh}	Electron-hole generation	Generation rate of ordinary mobile electron-hole pairs in a semiconductor or dielectric.
R_{eh}	Electron-hole recombination	Recombination rate of ordinary mobile electron-hole pairs.
Δ	Superconducting gap	Minimum energy of one quasiparticle excitation.
T_c	Critical temperature	Temperature below which superconductivity appears.
n_s	Superfluid density	Density participating in condensate response.
n_{cp}	Cooper-pair density	Number density of Cooper pairs; electron superfluid density is approximately $2n_{cp}$.

Symbol / acronym	Name	Meaning in this document
θ	Condensate phase	Phase of the superconducting order parameter.
$\mathbf{A}$	Magnetic vector potential	Vector potential entering gauge-invariant condensate momentum.
$\mathbf{J}_s$	Supercurrent density	Current density carried by the superconducting condensate.
$\mathbf{J}_{qp}$	Quasiparticle current density	Current density carried by quasiparticle excitations.
$\mathbf{J}_{tot}$	Total current density	Sum of condensate, quasiparticle, and possibly displacement or normal currents.
n_{qp}	Quasiparticle density	Number of QPs per unit superconducting-film volume.
x_{qp}	Normalized QP density	Common dimensionless measure, roughly $n_{qp}/(2N_0\Delta)$.
N_0	Single-spin density of states	Normal-state density of states at the Fermi level.
D_{qp}	QP diffusivity	Effective diffusion coefficient for QPs in a superconducting film.
G_{qp}	QP generation rate	Rate at which QPs are produced by pair-breaking phonons, photons, radiation, or injection.
R_{qp}	QP recombination coefficient	Coefficient for approximate QP recombination loss.
τ_{qp}	QP lifetime	Effective time scale for QP removal by recombination, trapping, or escape.
τ_{trap}	Trap time	Characteristic time for QP trapping/removal.
Γ_{qp}	QP tunneling rate	Rate of QP tunneling across a Josephson junction or weak link.
Γ_{pb}	Pair-breaking generation rate	QP generation rate caused by pair-breaking phonons.
μ_{chem}	Chemical potential	Thermodynamic chemical potential of particles.
μ_{ec}	Electrochemical potential	Chemical plus electrostatic potential energy.
μ^*	Charge-imbalance potential	Nonequilibrium quasiparticle charge-mode potential.

Symbol / acronym	Name	Meaning in this document
I_c	Critical current	Maximum Josephson supercurrent in the zero-voltage branch.
δ	Junction phase difference	Gauge-invariant phase difference across a Josephson junction.
I_J	Josephson current	$I_J = I_c \sin \delta$.
V_J	Junction voltage	Voltage across a Josephson junction.
C_J	Junction capacitance	Capacitance across the tunnel barrier.
E_J	Josephson energy	$E_J = \hbar I_c / (2e)$.
E_C	Charging energy	Island charging scale, often $E_C = e^2 / (2C_\Sigma)$.
Q_{off}	Offset charge	Effective environmental charge coupled capacitively to a superconducting island.
n_g	Dimensionless gate charge	$n_g = Q_g / (2e)$, with radiation/traps contributing to Q_g .
N_t	Trap density	Density of localized charge traps.
f_t	Trap occupation	Occupation probability of a trap.
ρ_{off}	Offset-charge density	Radiation-induced charge density that couples electrostatically to superconducting islands.
ψ	Weighting potential	Solution of an auxiliary electrostatic problem used to compute induced island charge.
T_{ph}	Phonon temperature	Effective phonon-bath temperature from Module 4, when local equilibrium is meaningful.
T_e	Electron temperature	Effective electronic temperature, if the electronic subsystem is thermalized.

1 What Module 5 did before superconductivity

Baseline Module 5 is a semiconductor drift-diffusion carrier-transport module. Its unknowns are mobile electron and hole concentrations,

$$n(\mathbf{x}, t), \quad p(\mathbf{x}, t), \quad (1.1)$$

which are driven by the electrostatic potential ϕ , electric field $\mathbf{E} = -\nabla\phi$, temperature T , and defect populations C_j provided by Modules 2, 3, and 4. In a two-dimensional reduced model, the usual

continuity equations are

$$\frac{\partial n}{\partial t} + \nabla \cdot \mathbf{\Gamma}_n = G_n - R_n, \quad (1.2)$$

$$\frac{\partial p}{\partial t} + \nabla \cdot \mathbf{\Gamma}_p = G_p - R_p, \quad (1.3)$$

where $\mathbf{\Gamma}_n$ and $\mathbf{\Gamma}_p$ are particle fluxes. The conventional electrical currents are

$$\mathbf{J}_n = q\mu_n n \mathbf{E} + qD_n \nabla n, \quad (1.4)$$

$$\mathbf{J}_p = q\mu_p p \mathbf{E} - qD_p \nabla p, \quad (1.5)$$

$$\mathbf{J} = \mathbf{J}_n + \mathbf{J}_p. \quad (1.6)$$

Defects enter through mobility reduction, generation/recombination, and space-charge feedback:

$$\mu_n = \mu_{n0}(T)F_n(C_1, C_2, \dots), \quad (1.7)$$

$$\mu_p = \mu_{p0}(T)F_p(C_1, C_2, \dots), \quad (1.8)$$

$$R \supset R_{\text{SRH}}(n, p, N_t, E_t, T), \quad (1.9)$$

$$\rho = q(p - n + N_D^+ - N_A^- + \rho_{\text{def}}/q). \quad (1.10)$$

Thus the baseline causal chain is

$$\boxed{\phi, \mathbf{E}, T, C_j \longrightarrow \mu_n, \mu_p, G, R \longrightarrow n, p, \mathbf{J} \longrightarrow y_{\text{device}}.} \quad (1.11)$$

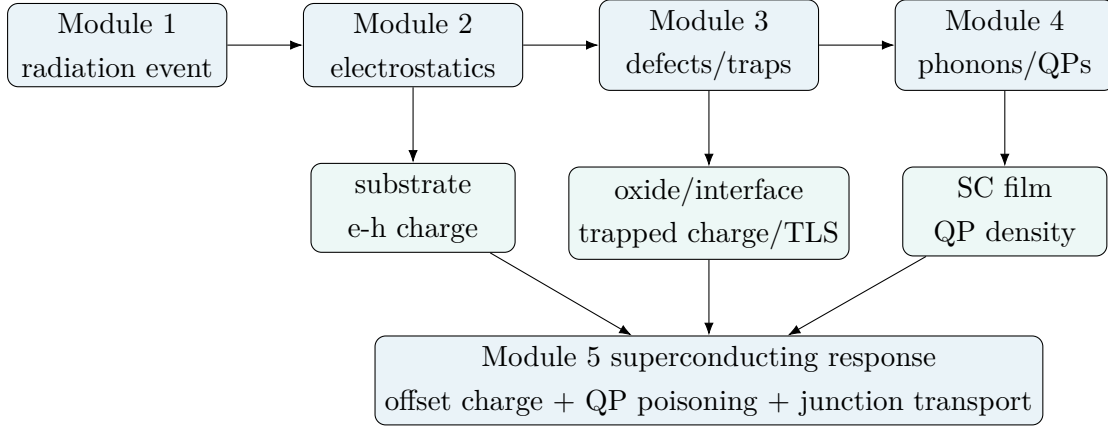
Baseline Module 5 intuition. Radiation changes semiconductor transport because it creates mobile carriers, creates traps, changes recombination, changes mobility, and changes the electric field. That picture is correct for ordinary semiconductor devices. It is not sufficient for a superconducting qubit or superconducting sensor because the electrically relevant carriers include a condensate and quasiparticles, not merely ordinary electrons and holes.

2 The first conceptual change below the superconducting transition

A superconducting device is a hybrid object. It may contain a silicon or sapphire substrate, oxide or dielectric layers, native interfaces, normal-metal structures, superconducting films, and Josephson junction tunnel barriers. Cooling the device below T_c does not make every region superconducting. Instead, the material regions must be separated:

$$\Omega = \Omega_{\text{sub}} \cup \Omega_{\text{ox}} \cup \Omega_{\text{sc}} \cup \Omega_{\text{JJ}} \cup \Omega_{\text{normal}}. \quad (2.1)$$

Each region carries charge differently.



The most important warning is this:

Do not replace n, p by one superconducting carrier density.

(2.2)

The superconducting condensate, quasiparticles, ordinary semiconductor carriers, trapped charges, and tunnel-barrier defects are different physical populations with different time scales and different equations.

2.1 Which carrier populations remain relevant?

In the superconducting extension of Module 5, the old variables n, p are not simply deleted. Instead, they become material-dependent and event-dependent:

$$n_{\text{sub}}(\mathbf{x}, t), p_{\text{sub}}(\mathbf{x}, t) \quad \text{transient radiation-generated substrate carriers,} \quad (2.3)$$

$$f_t(\mathbf{x}, t) \quad \text{trap/TLS occupations in oxide, interfaces, and junction barriers,} \quad (2.4)$$

$$\theta(\mathbf{x}, t), n_s(\mathbf{x}, T) \quad \text{condensate phase and superfluid density,} \quad (2.5)$$

$$n_{\text{qp}}(\mathbf{x}, t) \quad \text{quasiparticle density in superconducting films,} \quad (2.6)$$

$$\delta(t) \quad \text{Josephson phase difference across a junction,} \quad (2.7)$$

$$Q_{\text{off}}(t) \quad \text{radiation-induced offset charge coupled to islands.} \quad (2.8)$$

For a superconducting qubit application, the primary Module 5 output may no longer be a semiconductor leakage current. It may be a set of poisoning and charge-noise metrics:

$n_{\text{qp}}(\mathbf{x}, t), \quad \Gamma_{\text{qp}}(t), \quad Q_{\text{off}}(t), \quad \delta\omega_q(t), \quad \Gamma_1^{\text{qp}}(t), \quad \text{parity-switching rate.}$

(2.9)

3 Semiconductor carriers at cryogenic temperature: freeze-out and radiation bursts

Start with the part of the device that is not superconducting: the substrate or dielectric. If the substrate is silicon at millikelvin temperature, thermally generated mobile carrier densities are usually strongly suppressed. A standard nondegenerate semiconductor estimate is

$$n_i(T) \propto T^{3/2} \exp \left[-\frac{E_g}{2k_B T} \right]. \quad (3.1)$$

At cryogenic temperature, this equilibrium density is negligible for a large-gap semiconductor. Donors and acceptors may also freeze out, because the thermal energy is too small to ionize dopants efficiently. Thus the baseline steady drift-diffusion background often collapses to

$$n_{\text{eq}} \approx 0, \quad p_{\text{eq}} \approx 0 \quad (3.2)$$

for an insulating substrate or depleted semiconductor region.

But radiation changes this. A single ionizing event can create many electron-hole pairs and high-energy phonons. Therefore the correct cryogenic substrate equation is not a steady room-temperature drift-diffusion problem. It is an event-driven transient transport and trapping problem:

$$\frac{\partial n_{\text{sub}}}{\partial t} + \nabla \cdot \mathbf{\Gamma}_{n,\text{sub}} = G_{eh}^{\text{rad}} - R_{eh}^{\text{sub}} - R_{n \rightarrow t}, \quad (3.3)$$

$$\frac{\partial p_{\text{sub}}}{\partial t} + \nabla \cdot \mathbf{\Gamma}_{p,\text{sub}} = G_{eh}^{\text{rad}} - R_{eh}^{\text{sub}} - R_{p \rightarrow t}, \quad (3.4)$$

$$\frac{\partial f_t}{\partial t} = c_n n_{\text{sub}} (1 - f_t) - e_n f_t - c_p p_{\text{sub}} f_t + e_p (1 - f_t). \quad (3.5)$$

Here f_t is a trap occupation, c_n, c_p are capture coefficients, and e_n, e_p are emission coefficients. At millikelvin temperature, thermally activated emission may be extremely slow. In reduced Module 5 language, this means trap occupancy can store the memory of a radiation event.

3.1 Drift and diffusion are still possible, but the parameters change

For transient radiation-generated substrate charge, the old drift-diffusion flux forms can still be used as a reduced model:

$$\mathbf{\Gamma}_{n,\text{sub}} = -\mu_n n_{\text{sub}} \mathbf{E} - D_n \nabla n_{\text{sub}}, \quad (3.6)$$

$$\mathbf{\Gamma}_{p,\text{sub}} = +\mu_p p_{\text{sub}} \mathbf{E} - D_p \nabla p_{\text{sub}}. \quad (3.7)$$

However, the cryogenic parameter regime is different:

$$\mu_{n,p}(T, C_j) \quad \text{may increase from reduced phonon scattering,} \quad (3.8)$$

$$\mu_{n,p}(T, C_j) \quad \text{may decrease from impurity, interface, and defect scattering,} \quad (3.9)$$

$$D_{n,p} = \mu_{n,p} \frac{k_B T_{\text{carrier}}}{q} \quad \text{only when carriers are locally thermalized.} \quad (3.10)$$

The Einstein relation is not automatic after an energetic radiation event. If the carrier distribution is nonthermal, the diffusion coefficient should be treated as an effective transport coefficient or derived from a kinetic calculation.

Modeling decision for this project. For a first superconducting Module 5 extension, it is reasonable to keep a reduced transient drift-diffusion equation in the substrate/dielectric only as a charge-collection and trapped-charge model. It should not be interpreted as ordinary steady semiconductor device operation at room temperature.

4 Superconducting current is not ordinary drift-diffusion

In a superconducting film, the low-energy charged fluid is the condensate. A useful order parameter is

$$\Psi(\mathbf{x}, t) = \sqrt{n_{\text{cp}}(\mathbf{x}, t)} e^{i\theta(\mathbf{x}, t)}, \quad (4.1)$$

where n_{cp} is Cooper-pair density and θ is phase. The gauge-invariant condensate momentum is

$$\mathbf{p}_s = \hbar \nabla \theta - 2e \mathbf{A}. \quad (4.2)$$

The condensate velocity is

$$\mathbf{v}_s = \frac{1}{2m^*} (\hbar \nabla \theta - 2e \mathbf{A}), \quad (4.3)$$

where $2m^*$ is the approximate Cooper-pair mass. Since each Cooper pair has charge $-2e$, the supercurrent density is

$$\mathbf{J}_s = -2en_{\text{cp}} \mathbf{v}_s = -\frac{en_{\text{cp}}}{m^*} (\hbar \nabla \theta - 2e \mathbf{A}). \quad (4.4)$$

Depending on sign convention for electron charge and phase, the sign may be written differently. The invariant statement is that the supercurrent is proportional to the gauge-invariant phase gradient.

This is not a drift-diffusion law. It does not say

$$\mathbf{J}_s = \sigma \mathbf{E} - qD \nabla n. \quad (4.5)$$

In an ideal DC superconducting state, a static electric field inside a bulk superconductor is screened and the electrochemical potential is constant. Sustained DC current is carried by phase stiffness,

not by resistive drift in an electric field.

4.1 Continuity equation for condensate charge

The condensate charge density is approximately

$$\rho_s = -2en_{\text{cp}}. \quad (4.6)$$

If condensate density changes, charge conservation gives

$$\frac{\partial \rho_s}{\partial t} + \nabla \cdot \mathbf{J}_s = S_{\text{pair}}, \quad (4.7)$$

where S_{pair} represents conversion between condensate pairs and quasiparticles. In equilibrium and far from pair breaking, $S_{\text{pair}} \approx 0$ and n_{cp} is effectively fixed by temperature and material properties. For most Module 5 superconducting-device calculations, one should not evolve n_{cp} as a free drift-diffusion carrier density. Instead, use material properties $\Delta(T, C_j)$, $T_c(C_j)$, kinetic inductance, and junction parameters.

4.2 Kinetic inductance as the transport output of the condensate

The condensate has inertia. A changing current therefore stores kinetic energy. The kinetic-energy density of the condensate is

$$u_{\text{kin}} = \frac{1}{2}(2m^*)n_{\text{cp}}v_s^2. \quad (4.8)$$

Because $\mathbf{J}_s = -2en_{\text{cp}}\mathbf{v}_s$,

$$u_{\text{kin}} = \frac{m^*}{4e^2n_{\text{cp}}}J_s^2. \quad (4.9)$$

This motivates a kinetic inductance. In a thin superconducting wire or film, reduced n_{cp} or increased disorder increases kinetic inductance. Radiation-induced disorder from Module 3 and QP population from Module 4 can therefore feed Module 5 by changing inductive response even without ordinary DC resistance.

5 Quasiparticle transport in superconducting films

A quasiparticle is an excitation above the superconducting ground state. It is not the same object as a normal-state electron, but at the reduced modeling level one often evolves the quasiparticle density n_{qp} . The simplest spatial equation is a diffusion-reaction equation,

$$\frac{\partial n_{\text{qp}}}{\partial t} = \nabla \cdot (D_{\text{qp}} \nabla n_{\text{qp}}) + G_{\text{qp}} - R_{\text{qp}} n_{\text{qp}}^2 - \frac{n_{\text{qp}} - n_{\text{qp}}^{\text{eq}}}{\tau_{\text{trap}}} + S_{\text{inj}}. \quad (5.1)$$

The terms have direct physical meanings:

$$\nabla \cdot (D_{\text{qp}} \nabla n_{\text{qp}}) \quad \text{spatial spreading of QPs in the superconducting film,} \quad (5.2)$$

$$G_{\text{qp}} \quad \text{generation by pair-breaking phonons, photons, radiation, or injection,} \quad (5.3)$$

$$R_{\text{qp}} n_{\text{qp}}^2 \quad \text{two-QP recombination into a Cooper pair plus phonon energy,} \quad (5.4)$$

$$(n_{\text{qp}} - n_{\text{qp}}^{\text{eq}})/\tau_{\text{trap}} \quad \text{removal by traps, gap-engineered sinks, or escape,} \quad (5.5)$$

$$S_{\text{inj}} \quad \text{intentional or parasitic injection from leads or junctions.} \quad (5.6)$$

The equilibrium density at low temperature is exponentially small. A common BCS low-temperature estimate is

$$n_{\text{qp}}^{\text{eq}}(T) \approx 2N_0 \sqrt{2\pi k_B T \Delta} \exp\left(-\frac{\Delta}{k_B T}\right), \quad (5.7)$$

where N_0 is the single-spin normal-state density of states at the Fermi level. This exponential factor is why even small nonequilibrium sources can dominate over equilibrium QPs.

5.1 Why quasiparticle transport is Module 5, not only Module 4

Module 4 treats energy flow: phonons, pair breaking, recombination energy, and heat. Module 5 treats electrical response. Quasiparticles are an overlap variable because they are both excitations carrying energy and mobile charged degrees of freedom that affect device response. The clean separation is

$$\text{Module 4: where does deposited energy go?} \quad (5.8)$$

$$\text{Module 5: how do the resulting QPs and charges change electrical response?} \quad (5.9)$$

Thus n_{qp} may be computed in Module 4 and then consumed in Module 5, or it may be part of a coupled Module 5 response model. In either case, Module 5 needs n_{qp} to compute junction poisoning, parity switching, conductance, and qubit relaxation.

5.2 Normalized quasiparticle density

A useful dimensionless QP density is

$$x_{\text{qp}} = \frac{n_{\text{qp}}}{2N_0 \Delta}. \quad (5.10)$$

The denominator is approximately the density of Cooper-pair-scale electronic states near the gap. In superconducting-qubit literature, small x_{qp} can still produce measurable relaxation and parity switching because junction tunneling is sensitive to QPs near the gap edge.

6 Quasiparticle current and charge imbalance

Equation (5.1) tracks QP number, not necessarily charge current. In superconductors, quasiparticles are coherent mixtures of electron-like and hole-like excitations. A quasiparticle at energy E has BCS coherence amplitudes u_E and v_E , and its effective charge expectation is roughly

$$q_E = -e \left(u_E^2 - v_E^2 \right). \quad (6.1)$$

Near the gap edge, electron-like and hole-like character can nearly cancel. Therefore a QP density gradient does not automatically imply the same electrical current as an ordinary electron density gradient.

A reduced charge-imbalance description introduces a nonequilibrium quasiparticle chemical-potential-like variable μ^* . Then a linear transport form is

$$\mathbf{J}_{\text{qp}} \approx -\sigma_{\text{qp}} \nabla \left(\frac{\mu^*}{e} \right), \quad (6.2)$$

with a relaxation equation

$$\frac{\partial \mu^*}{\partial t} = D_Q \nabla^2 \mu^* - \frac{\mu^*}{\tau_Q} + S_Q. \quad (6.3)$$

Here τ_Q is a charge-imbalance relaxation time and S_Q represents injection or radiation-generated imbalance. This is a higher-fidelity transport model than simply equating QP density diffusion with electrical current.

Practical reduced model. For the first RadiationEffects_proj2 superconducting Module 5 extension, use n_{qp} as the main QP transport state and compute poisoning metrics from it. Add μ^* only if the project specifically needs nonequilibrium charge-imbalance voltages or quasiparticle charge current.

7 Josephson junction transport replaces ordinary drift current across the weak link

The most important electrical nonlinearity in a superconducting qubit is often the Josephson junction. A tunnel junction has two kinds of current:

$$I = I_s + I_{\text{qp}} + I_C + I_{\text{noise}}. \quad (7.1)$$

The supercurrent branch is

$$I_s = I_c \sin \delta, \quad (7.2)$$

where δ is the gauge-invariant phase difference. The phase and voltage are related by

$$V_J = \frac{\hbar}{2e} \frac{\partial \delta}{\partial t}. \quad (7.3)$$

The capacitive current is

$$I_C = C_J \frac{\partial V_J}{\partial t}. \quad (7.4)$$

A resistively and capacitively shunted reduced junction equation is therefore

$$C_J \frac{\hbar}{2e} \frac{\partial^2 \delta}{\partial t^2} + \frac{1}{R_{\text{qp}}} \frac{\hbar}{2e} \frac{\partial \delta}{\partial t} + I_c \sin \delta = I_{\text{bias}} + I_{\text{noise}}, \quad (7.5)$$

where R_{qp} should be interpreted carefully: it is not an ordinary room-temperature semiconductor resistance. It represents dissipative quasiparticle tunneling and any effective subgap conductance.

7.1 Josephson energy and charging energy

The Josephson energy follows from integrating the supercurrent with respect to phase:

$$E_J(\delta) = -E_J \cos \delta, \quad E_J = \frac{\hbar I_c}{2e}. \quad (7.6)$$

A superconducting island also stores electrostatic energy. If its total capacitance is C_Σ , the charging energy scale is

$$E_C = \frac{e^2}{2C_\Sigma}. \quad (7.7)$$

For an island with integer Cooper-pair number N and dimensionless offset charge n_g , the charging term is

$$H_C = 4E_C(N - n_g)^2, \quad n_g = \frac{Q_g + Q_{\text{off}}}{2e}. \quad (7.8)$$

This equation is central to the radiation problem. Radiation-induced trapped charge does not have to conduct through the junction to affect the qubit. It can shift Q_{off} , thereby shifting n_g , thereby changing device frequency or parity-sensitive observables.

8 Quasiparticle tunneling and poisoning

Quasiparticles in a superconducting electrode can tunnel across a Josephson junction. This tunneling can change the island parity, dissipate energy, and relax a qubit. A reduced local model writes the QP tunneling rate as proportional to the QP density near the junction:

$$\Gamma_{\text{qp}}(t) \approx \gamma_{\text{tun}} \overline{x_{\text{qpJJ}}}(t), \quad (8.1)$$

where $\overline{x_{qpJJ}}$ is a junction-neighborhood average of x_{qp} , and γ_{tun} is a calibrated coefficient depending on junction conductance, gap, matrix elements, and qubit mode. A more explicit averaging definition is

$$\overline{x_{qpJJ}}(t) = \frac{\int_{\Omega_{sc}} W_{JJ}(\mathbf{x}) x_{qp}(\mathbf{x}, t) d\Omega}{\int_{\Omega_{sc}} W_{JJ}(\mathbf{x}) d\Omega}, \quad (8.2)$$

where W_{JJ} weights the electrode regions that efficiently tunnel into the junction.

For qubit relaxation modeling, a reduced proportionality is often sufficient:

$$\Gamma_1^{qp}(t) \equiv \frac{1}{T_{1,qp}(t)} \approx A_q \overline{x_{qpJJ}}(t), \quad (8.3)$$

where A_q is a device-dependent coefficient. This is not a substitute for microscopic tunneling theory, but it is the correct reduced response architecture for RadiationEffects_proj2: radiation source $\rightarrow$ phonons/QPs $\rightarrow$ local QP density near junction $\rightarrow$ device-level degradation metric.

8.1 Pair-breaking generation source from Module 4

Module 4 supplies a pair-breaking phonon field or a reduced pair-breaking source. In Module 5 this enters as

$$G_{qp}(\mathbf{x}, t) = \Gamma_{pb}(\mathbf{x}, t) + G_{IR}(\mathbf{x}, t) + G_{inj}(\mathbf{x}, t), \quad (8.4)$$

where Γ_{pb} is due to high-energy phonons with $\hbar\omega \geq 2\Delta$, G_{IR} is due to stray photons/infrared radiation, and G_{inj} represents junction or lead injection.

A reduced energy-to-QP conversion relation is

$$N_{qp,gen} \approx \eta_{qp} \frac{E_{abs,sc}}{\Delta}, \quad (8.5)$$

where $E_{abs,sc}$ is radiation-induced energy absorbed into the superconducting film and η_{qp} is an efficiency factor. The factor is not universal because some energy downconverts into subgap phonons or escapes before breaking pairs.

9 Offset charge from radiation-generated carriers and traps

Radiation can create electron-hole pairs in the substrate and trapped charge in dielectrics or interfaces. Even if this charge never becomes a conduction current, it can capacitively shift a superconducting island. Module 2 computes electrostatics; Module 3 computes trap populations; Module 5 converts them into device response.

Let $\rho_{off}(\mathbf{x}, t)$ be the radiation-induced charge density outside the superconducting island. The induced charge on island a can be computed using a weighting potential ψ_a . Define ψ_a by the

auxiliary electrostatic problem

$$\nabla \cdot (\epsilon \nabla \psi_a) = 0 \quad \text{in dielectric/substrate regions,} \quad (9.1)$$

$$\psi_a = 1 \quad \text{on island } a, \quad (9.2)$$

$$\psi_a = 0 \quad \text{on other conductors,} \quad (9.3)$$

$$\mathbf{n} \cdot \epsilon \nabla \psi_a = 0 \quad \text{on insulating outer boundaries, if appropriate.} \quad (9.4)$$

Then the induced offset charge is

$$Q_{\text{off}, a}(t) = \int_{\Omega \setminus \Omega_{\text{metal}}} \rho_{\text{off}}(\mathbf{x}, t) \psi_a(\mathbf{x}) d\Omega. \quad (9.5)$$

This is the carrier-transport consequence of radiation-generated charge. The charge distribution comes from substrate carrier motion and trap occupation; the device response comes from the induced Q_{off} .

9.1 Trap occupation as a stochastic or continuum variable

For a trap species with charge q_t , density $N_t(\mathbf{x})$, and occupation $f_t(\mathbf{x}, t)$, the trapped-charge density is

$$\rho_t(\mathbf{x}, t) = q_t N_t(\mathbf{x}) f_t(\mathbf{x}, t). \quad (9.6)$$

Substitution into Eq. (9.5) gives

$$Q_{\text{off}, a}(t) = \sum_t \int q_t N_t^{(t)}(\mathbf{x}) f_t^{(t)}(\mathbf{x}, t) \psi_a(\mathbf{x}) d\Omega. \quad (9.7)$$

For a single microscopic trap switching between two occupations, the contribution is a random telegraph signal,

$$\Delta Q_a = q_t \psi_a(\mathbf{x}_t), \quad (9.8)$$

where $\mathbf{x}_t$ is the trap location. This is the bridge between Module 3 TLS/trap physics and Module 5 charge-noise response.

Key intuition. In a superconducting qubit, a radiation-created charge does not need to become a measured current to matter. It can simply move or become trapped near a high-participation electrostatic region, changing offset charge and therefore changing the qubit Hamiltonian.

10 Capacitance-matrix formulation for superconducting islands

A superconducting circuit with multiple islands is naturally described by a capacitance matrix. If island voltages are collected in $\mathbf{V}$, the island charges satisfy

$$\mathbf{Q} = \mathbf{C}\mathbf{V} + \mathbf{Q}_{\text{off}}. \quad (10.1)$$

The electrostatic energy is

$$U_C = \frac{1}{2} (\mathbf{Q} - \mathbf{Q}_{\text{off}})^T \mathbf{C}^{-1} (\mathbf{Q} - \mathbf{Q}_{\text{off}}). \quad (10.2)$$

For a single island, this reduces to the familiar charging-energy expression. For a multi-island device, radiation-induced charge can shift several islands with correlations determined by the weighting potentials and capacitance matrix.

This provides a clean Module 5 output map:

$$\boxed{\rho_{\text{off}}(\mathbf{x}, t) \longrightarrow \mathbf{Q}_{\text{off}}(t) \longrightarrow U_C(t) \longrightarrow \delta\omega_q(t), \delta n_g(t), \text{ charge-noise metric.}} \quad (10.3)$$

11 How defects alter superconducting carrier response

Module 3 supplies defect fields. In ordinary Module 5, those defects modify electron and hole mobility and SRH recombination. In the superconducting extension, defects modify a different set of response parameters:

$$C_j(\mathbf{x}, t) \longrightarrow \Delta(\mathbf{x}, t), T_c(\mathbf{x}, t), n_s(\mathbf{x}, t), \quad (11.1)$$

$$C_j(\mathbf{x}, t) \longrightarrow D_{\text{qp}}(\mathbf{x}, t), R_{\text{qp}}(\mathbf{x}, t), \tau_{\text{trap}}(\mathbf{x}, t), \quad (11.2)$$

$$C_j(\mathbf{x}, t) \longrightarrow I_c(t), R_N(t), C_J(t), \Gamma_{\text{qp}}(t), \quad (11.3)$$

$$C_j(\mathbf{x}, t) \longrightarrow N_t(\mathbf{x}, t), f_t(\mathbf{x}, t), Q_{\text{off}}(t). \quad (11.4)$$

Here R_N is the normal-state junction resistance or effective tunnel resistance. The exact maps are material-specific. A reduced phenomenological version may be

$$\Delta(\mathbf{x}, t) = \Delta_0 \left[1 - a_\Delta \sum_j C_j(\mathbf{x}, t) \right]_+, \quad (11.5)$$

$$D_{\text{qp}}(\mathbf{x}, t) = \frac{1}{3} v_{\text{qp}}(\Delta, T) \ell_{\text{qp}}(\mathbf{x}, t), \quad (11.6)$$

$$\ell_{\text{qp}}^{-1}(\mathbf{x}, t) = \ell_0^{-1} + \sum_j \alpha_j C_j(\mathbf{x}, t), \quad (11.7)$$

$$\tau_{\text{trap}}^{-1}(\mathbf{x}, t) = \tau_0^{-1} + \sum_j \beta_j C_j(\mathbf{x}, t). \quad (11.8)$$

These equations should be treated as reduced closure laws, not universal constants. Their value is that they preserve the causal structure of the multiphysics model.

12 Radiation event chain for superconducting Module 5

For a superconducting qubit-like device, a useful event chain is

$$\boxed{E_{\text{dep}} \rightarrow \{e\text{-}h \text{ pairs, phonons, traps}\} \rightarrow \{Q_{\text{off}}, n_{\text{qp}}\} \rightarrow \{\Gamma_1^{\text{qp}}, \Gamma_{\text{parity}}, \delta\omega_q\}.} \quad (12.1)$$

This chain has two main branches.

12.1 Charge branch

The charge branch is

$$E_{\text{dep}} \longrightarrow G_{eh}^{\text{rad}}(\mathbf{x}, t), \quad (12.2)$$

$$G_{eh}^{\text{rad}} \longrightarrow n_{\text{sub}}, p_{\text{sub}}, f_{\text{t}}, \quad (12.3)$$

$$n_{\text{sub}}, p_{\text{sub}}, f_{\text{t}} \longrightarrow \rho_{\text{off}}(\mathbf{x}, t), \quad (12.4)$$

$$\rho_{\text{off}} \longrightarrow Q_{\text{off}}(t), \quad (12.5)$$

$$Q_{\text{off}} \longrightarrow n_{\text{g}}(t), \delta\omega_q(t), \text{ charge-noise observables.} \quad (12.6)$$

A reduced generation estimate is

$$G_{eh}^{\text{rad}}(\mathbf{x}, t) = \eta_{eh} \frac{\dot{q}_{\text{dep}}(\mathbf{x}, t)}{\varepsilon_{eh}}, \quad (12.7)$$

where ε_{eh} is the mean energy per electron-hole pair in the substrate material and η_{eh} is an efficiency factor.

12.2 Quasiparticle branch

The quasiparticle branch is

$$E_{\text{dep}} \longrightarrow \text{high-energy phonons}, \quad (12.8)$$

$$\text{high-energy phonons} \longrightarrow \Gamma_{\text{pb}}(\mathbf{x}, t), \quad (12.9)$$

$$\Gamma_{\text{pb}} \longrightarrow n_{\text{qp}}(\mathbf{x}, t), \quad (12.10)$$

$$n_{\text{qp}} \longrightarrow \Gamma_{\text{qp}}(t), \Gamma_1^{\text{qp}}(t), \text{ parity switching.} \quad (12.11)$$

A reduced QP equation with Module 4 source input is

$$\frac{\partial n_{\text{qp}}}{\partial t} = \nabla \cdot (D_{\text{qp}} \nabla n_{\text{qp}}) + \eta_{\text{pb}} \frac{P_{2\Delta}(\mathbf{x}, t)}{\Delta} - R_{\text{qp}} n_{\text{qp}}^2 - \frac{n_{\text{qp}}}{\tau_{\text{trap}}}, \quad (12.12)$$

where $P_{2\Delta}$ is the power density in phonons energetic enough to break pairs.

13 Boundary and interface conditions

Superconducting Module 5 needs different boundary conditions for each carrier type.

13.1 Substrate charge boundaries

For transient substrate carriers,

$$\mathbf{\Gamma}_{n,p} \cdot \mathbf{n} = 0 \quad (13.1)$$

represents insulating boundaries. A collecting electrode or recombination surface can be modeled by

$$\mathbf{\Gamma}_n \cdot \mathbf{n} = S_n(n - n_{\text{eq}}), \quad \mathbf{\Gamma}_p \cdot \mathbf{n} = S_p(p - p_{\text{eq}}), \quad (13.2)$$

where S_n, S_p are surface loss velocities. At cryogenic temperature these should often be interpreted as trapping/collection rates, not thermal equilibrium recombination speeds.

13.2 Quasiparticle film boundaries

For QP diffusion in a superconducting film,

$$-D_{\text{qp}} \nabla n_{\text{qp}} \cdot \mathbf{n} = 0 \quad (13.3)$$

represents reflecting boundaries. A QP trap or sink can be modeled by

$$-D_{\text{qp}} \nabla n_{\text{qp}} \cdot \mathbf{n} = S_{\text{qp}} (n_{\text{qp}} - n_{\text{qp}}^{\text{trap}}). \quad (13.4)$$

At a material interface, continuity of QP current and appropriate transmission/trapping conditions are needed:

$$n_{\text{qp}}^{(1)} \leftrightarrow n_{\text{qp}}^{(2)} \quad \text{through an interface conductance or transmission law,} \quad (13.5)$$

$$-D_1 \nabla n_{\text{qp}}^{(1)} \cdot \mathbf{n}_1 = G_{12} (n_{\text{qp}}^{(1)} - \alpha_{12} n_{\text{qp}}^{(2)}). \quad (13.6)$$

13.3 Junction coupling boundary

Near a Josephson junction, QP density should feed a local tunneling sink/source:

$$-D_L \nabla n_{qp}^L \cdot \mathbf{n} = S_{\text{tun}} (n_{qp}^L - n_{qp}^R) + S_{\text{relax}} n_{qp}^L, \quad (13.7)$$

with a corresponding equation on the right electrode. In a reduced model, this can be replaced by computing $\overline{x_{qpJJ}}$ and using Eq. (8.1) without explicitly draining QPs through the junction.

14 FEM weak forms for superconducting Module 5

The existing Module 5 FEM path uses weak forms for drift-diffusion of n and p . In the superconducting extension, the simplest new FEM equation is the QP diffusion-reaction equation.

Let w be a test function. Starting from

$$\frac{\partial n_{qp}}{\partial t} - \nabla \cdot (D_{qp} \nabla n_{qp}) = G_{qp} - R_{qp} n_{qp}^2 - \frac{n_{qp}}{\tau_{\text{trap}}}, \quad (14.1)$$

multiply by w and integrate:

$$\int_{\Omega_{\text{sc}}} w \frac{\partial n_{qp}}{\partial t} d\Omega + \int_{\Omega_{\text{sc}}} D_{qp} \nabla w \cdot \nabla n_{qp} d\Omega - \int_{\partial\Omega_{\text{sc}}} w D_{qp} \nabla n_{qp} \cdot \mathbf{n} d\Gamma = \int_{\Omega_{\text{sc}}} w \left(G_{qp} - R_{qp} n_{qp}^2 - \frac{n_{qp}}{\tau_{\text{trap}}} \right) d\Omega. \quad (14.2)$$

For zero-flux boundaries, the boundary term vanishes. With finite elements

$$n_{qp}(\mathbf{x}, t) \approx \sum_a N_a(\mathbf{x}) \tilde{n}_{qp,a}(t), \quad (14.3)$$

the semi-discrete equation is

$$\mathbf{M} \dot{\tilde{\mathbf{n}}}_{qp} + \mathbf{K}_{qp} \tilde{\mathbf{n}}_{qp} = \mathbf{f}_{\text{gen}} - \mathbf{f}_{\text{rec}}(\tilde{\mathbf{n}}_{qp}) - \mathbf{K}_{\text{trap}} \tilde{\mathbf{n}}_{qp}. \quad (14.4)$$

The element matrices are

$$M_{ab}^{(e)} = \int_{\Omega_e} N_a N_b d\Omega, \quad (14.5)$$

$$K_{ab,qp}^{(e)} = \int_{\Omega_e} D_{qp} \nabla N_a \cdot \nabla N_b d\Omega, \quad (14.6)$$

$$(K_{\text{trap}}^{(e)})_{ab} = \int_{\Omega_e} \frac{1}{\tau_{\text{trap}}} N_a N_b d\Omega, \quad (14.7)$$

$$(f_{\text{gen}}^{(e)})_a = \int_{\Omega_e} N_a G_{qp} d\Omega. \quad (14.8)$$

A backward-Euler time step gives

$$\left(\frac{M}{\Delta t} + \mathbf{K}_{\text{qp}} + \mathbf{K}_{\text{trap}} + \mathbf{K}_{\text{rec}}^{\text{lin}} \right) \tilde{n}_{\text{qp}}^{k+1} = \frac{M}{\Delta t} \tilde{n}_{\text{qp}}^k + \mathbf{f}_{\text{gen}}^{k+1} + \mathbf{b}_{\text{rec}}^{\text{lin}}. \quad (14.9)$$

The nonlinear recombination term $R_{\text{qp}} n_{\text{qp}}^2$ can be handled explicitly, semi-implicitly, or by Newton iteration.

14.1 Weak form for trap occupation

If traps are modeled as a continuum field f_t with no spatial diffusion,

$$\frac{\partial f_t}{\partial t} = F(f_t, n, p, T), \quad (14.10)$$

then the weak form is simply

$$\int_{\Omega_t} w \frac{\partial f_t}{\partial t} d\Omega = \int_{\Omega_t} w F(f_t, n, p, T) d\Omega. \quad (14.11)$$

This gives a local reaction update at each node or quadrature point. If trap hopping or TLS diffusion is included, a diffusion term may be added, but it should not be assumed by default.

15 PINN and causal-PINN implications

The baseline Module 5 PINN would learn $n(x, y, t)$, $p(x, y, t)$, or current response subject to drift-diffusion residuals. The superconducting Module 5 PINN must learn a different state vector:

$$\mathcal{U}_{5,\text{SC}}(\mathbf{x}, t) = \{n_{\text{sub}}, p_{\text{sub}}, f_t, n_{\text{qp}}, Q_{\text{off}}, \Gamma_{\text{qp}}, \Gamma_1^{\text{qp}}\}. \quad (15.1)$$

A reduced PINN loss could include

$$\mathcal{L}_{\text{sub}} = \left\| \frac{\partial n_{\text{sub}}}{\partial t} + \nabla \cdot \mathbf{\Gamma}_{n,\text{sub}} - G_{eh} + R_n \right\|^2 + \left\| \frac{\partial p_{\text{sub}}}{\partial t} + \nabla \cdot \mathbf{\Gamma}_{p,\text{sub}} - G_{eh} + R_p \right\|^2, \quad (15.2)$$

$$\mathcal{L}_{\text{trap}} = \left\| \frac{\partial f_t}{\partial t} - F(f_t, n, p, T) \right\|^2, \quad (15.3)$$

$$\mathcal{L}_{\text{qp}} = \left\| \frac{\partial n_{\text{qp}}}{\partial t} - \nabla \cdot (D_{\text{qp}} \nabla n_{\text{qp}}) - G_{\text{qp}} + R_{\text{qp}} n_{\text{qp}}^2 + \frac{n_{\text{qp}}}{\tau_{\text{trap}}} \right\|^2, \quad (15.4)$$

$$\mathcal{L}_{\text{offset}} = \left\| Q_{\text{off}}(t) - \int \rho_{\text{off}}(\mathbf{x}, t) \psi(\mathbf{x}) d\Omega \right\|^2, \quad (15.5)$$

$$\mathcal{L}_{\text{response}} = \left\| \Gamma_1^{\text{qp}}(t) - A_q \overline{x_{\text{qpJJ}}}(t) \right\|^2. \quad (15.6)$$

For a causal PINN, the natural ordering variable is real time after the radiation impact, or

alternatively radiation dose/cooldown step. The causal weights should enforce early event physics before late-time tails:

$$\mathcal{L}_{\text{CPINN}} = \sum_{k=1}^{N_t} w_k \mathcal{L}(t_k), \quad w_k = \exp \left[-\epsilon \sum_{j < k} \mathcal{L}(t_j) \right]. \quad (15.7)$$

This is more physically natural for Module 5 than for Module 2 because Module 5 is genuinely time-evolutionary.

Important difference from Module 2 CPINN. Module 2 Poisson electrostatics is elliptic, so causal ordering had to be introduced through continuation variables such as dose, loading, or coupling iteration. Module 5 superconducting response is naturally causal in time: radiation event, charge drift/trapping, phonon propagation, QP generation, QP diffusion, poisoning, recovery.

16 Suggested state vector for a future MATLAB implementation

A practical first implementation should not try to solve every microscopic superconducting transport equation. It should add a reduced superconducting response layer to the existing Module 5 architecture.

16.1 Minimum useful superconducting Module 5 state

The minimum state for superconducting-device relevance is

$$\mathcal{S}_5 = \{n_{\text{qp}}(\mathbf{x}, t), Q_{\text{off}}, a(t), \Gamma_1^{\text{qp}}(t), \Gamma_{\text{parity}}(t), \delta\omega_q(t)\}. \quad (16.1)$$

Here n_{qp} can be spatially resolved over superconducting films. The offset charge Q_{off}, a can be computed using weighting potentials from Module 2 and trap/charge distributions from Module 3. Device response metrics can be reduced scalar outputs.

16.2 Recommended MATLAB data fields

A future MATLAB structure could contain

$$\text{state.nqp} \quad \text{nodal QP density in superconducting-film mesh,} \quad (16.2)$$

$$\text{state.xqp} \quad \text{normalized QP density,} \quad (16.3)$$

$$\text{state.Qoff} \quad \text{island offset-charge vector,} \quad (16.4)$$

$$\text{state.trapOcc} \quad \text{trap occupation field,} \quad (16.5)$$

$$\text{state.GammaQP} \quad \text{junction QP tunneling or parity-switching rate,} \quad (16.6)$$

$$\text{state.Gamma1QP} \quad \text{QP-induced qubit relaxation rate,} \quad (16.7)$$

$$\text{material.Dqp} \quad \text{QP diffusivity field,} \quad (16.8)$$

$$\text{material.Delta} \quad \text{superconducting gap field,} \quad (16.9)$$

$$\text{material.tauTrap} \quad \text{QP trapping time field,} \quad (16.10)$$

$$\text{coupling.weightingPotential} \quad \text{Module 2 island weighting potentials.} \quad (16.11)$$

16.3 Reduced solver sequence

A first executable superconducting Module 5 solver can follow this sequence:

1. Read Module 2 electrostatic fields and island weighting potentials.
2. Read Module 3 trap/defect fields and convert them into ρ_{off} , D_{qp} , τ_{trap} , and Δ modifiers.
3. Read Module 4 pair-breaking phonon or QP generation source G_{qp} .
4. Evolve n_{qp} with diffusion, generation, recombination, and trapping.
5. Compute x_{qp} and junction-local averages $\overline{x_{\text{qpJJ}}}$.
6. Compute Γ_1^{qp} , parity-switching rate, or an abstract poisoning score.
7. Compute Q_{off} from trapped charge using weighting potentials.
8. Return a superconducting response vector for Module 6 and Module 7.

17 Device-level response metrics

A useful superconducting Module 5 implementation should not stop at the internal fields. It should convert fields into device-level observables. The exact observable depends on whether the target device is a transmon, resonator, kinetic-inductance detector, superconducting nanowire, or a generic superconducting interconnect. Still, the same response logic appears repeatedly.

17.1 Quasiparticle poisoning score

A geometry-aware poisoning score can be defined as

$$S_{\text{qp}}(t) = \frac{\int_{\Omega_{\text{sc}}} W_{\text{active}}(\mathbf{x}) x_{\text{qp}}(\mathbf{x}, t) d\Omega}{\int_{\Omega_{\text{sc}}} W_{\text{active}}(\mathbf{x}) d\Omega}, \quad (17.1)$$

where W_{active} is large near Josephson junction electrodes, high-current-density regions, or resonator regions that dominate loss. This score is not a new fundamental law. It is a reduced device metric that lets Module 7 compare shielding, trapping, and layout designs.

17.2 Charge-noise and offset-charge score

Similarly, a charge-response score for island a can be defined by

$$S_{Q,a}(t) = \frac{|Q_{\text{off},a}(t) - Q_{\text{off},a}(0)|}{2e}. \quad (17.2)$$

For a charge-sensitive qubit, this score directly measures motion in dimensionless gate charge. For a transmon in the large E_J/E_C limit, offset-charge sensitivity is exponentially suppressed but not always irrelevant: charge-sensitive transmons, small islands, parity effects, and trap switching can still reveal or mediate radiation-induced charge dynamics.

17.3 Kinetic-inductance response

In a resonator or superconducting film, quasiparticles reduce the superfluid density and increase dissipation. A reduced kinetic-inductance perturbation can be written as

$$\frac{\delta L_k}{L_k} \approx -\frac{\delta n_s}{n_s}, \quad (17.3)$$

with δn_s related to pair breaking and QP population. A generic resonator-frequency perturbation is then

$$\frac{\delta f_r}{f_r} \approx -\frac{\alpha_k}{2} \frac{\delta L_k}{L_k}, \quad (17.4)$$

where α_k is a kinetic-inductance participation factor. This gives Module 5 another possible scalar output when the target device is a resonator rather than a qubit.

18 Reduced-model fidelity ladder

Because superconducting carrier response spans microscopic tunneling, mesoscale diffusion, and circuit response, the implementation should be staged. A useful ladder is:

1. **Level 0: scalar response map.** Use total deposited energy or total pair-breaking power to estimate a scalar poisoning score and offset-charge score.
2. **Level 1: spatial QP diffusion.** Evolve $n_{\text{qp}}(\mathbf{x}, t)$ in superconducting films with generation, recombination, trapping, and reflecting/sink boundaries.

3. **Level 2: substrate charge and trap dynamics.** Evolve transient radiation-generated $n_{\text{sub}}, p_{\text{sub}}$ and trap occupation f_t , then compute Q_{off} using weighting potentials.
4. **Level 3: junction-local tunneling response.** Convert junction-neighborhood x_{qp} into parity switching and relaxation rates using calibrated or microscopic tunneling coefficients.
5. **Level 4: circuit-level Hamiltonian update.** Insert $Q_{\text{off}}(t)$, $E_J(t)$, $E_C(t)$, and loss terms into a superconducting-circuit model.

For RadiationEffects_proj2, Level 1 plus the offset-charge part of Level 2 is the best first target. It is rich enough to capture the superconducting physics change, but still compatible with the existing MATLAB FEM framework.

19 Verification tests

A superconducting Module 5 implementation should pass several reduced tests before being coupled to the full project.

19.1 Test 1: zero-source equilibrium

With $G_{\text{qp}} = 0$, no trapped charge, reflecting boundaries, and small equilibrium QP density,

$$n_{\text{qp}}(t) \rightarrow n_{\text{qp}}^{\text{eq}}, \quad Q_{\text{off}} \rightarrow 0. \quad (19.1)$$

The solver should not invent QPs or offset charge.

19.2 Test 2: uniform QP generation with no diffusion

If $D_{\text{qp}} = 0$, $\tau_{\text{trap}} \rightarrow \infty$, and $G_{\text{qp}} = G_0$ is uniform, then

$$\frac{\partial n_{\text{qp}}}{\partial t} = G_0 - R_{\text{qp}} n_{\text{qp}}^2. \quad (19.2)$$

For $n_{\text{qp}}(0) = 0$, the analytic solution is

$$n_{\text{qp}}(t) = \sqrt{\frac{G_0}{R_{\text{qp}}}} \tanh \left(\sqrt{G_0 R_{\text{qp}}} t \right). \quad (19.3)$$

This is an excellent test of the nonlinear recombination update.

19.3 Test 3: diffusion from a localized QP burst

With recombination and trapping turned off, a localized initial n_{qp} packet should spread diffusively while conserving total QP number:

$$\frac{d}{dt} \int_{\Omega_{\text{sc}}} n_{\text{qp}} d\Omega = 0. \quad (19.4)$$

With trapping or recombination enabled, the total should decrease monotonically.

19.4 Test 4: weighting-potential offset charge

Place a single point charge q_t at position $\mathbf{x}_t$. The induced island offset charge should be

$$\Delta Q_a = q_t \psi_a(\mathbf{x}_t). \quad (19.5)$$

Moving the charge along an equipotential contour of ψ_a should leave ΔQ_a unchanged.

19.5 Test 5: junction-local poisoning metric

Set a known QP density profile and a known junction weighting function W_{JJ} . The code should reproduce

$$\overline{x_{\text{qpJJ}}} = \frac{\int W_{\text{JJ}} x_{\text{qp}} d\Omega}{\int W_{\text{JJ}} d\Omega} \quad (19.6)$$

and therefore $\Gamma_1^{\text{qp}} = A_q \overline{x_{\text{qpJJ}}}$.

20 How this changes the role of Module 5 in the full project

The superconducting version of Module 5 is not a replacement of the original semiconductor drift-diffusion module by a lower-temperature drift-diffusion module. It is a different response layer.

Feature	Ordinary Module 5	Superconducting Module 5
Mobile carriers	Electrons and holes	Substrate event carriers, trap occupations, condensate phase, QPs
Main current	Drift-diffusion current	Supercurrent, QP tunneling, effective poisoning response
Role of defects	Mobility degradation and SRH recombination	Gap/disorder changes, QP trapping, junction defects, offset charge/TLSs
Thermal coupling	Temperature modifies mobility and recombination	Pair-breaking phonons generate QPs; QP recombination returns energy to phonons
Electrostatic coupling	Space charge modifies Poisson field	Trapped charge shifts island offset charge and circuit Hamiltonian
Device output	Leakage current, conductance, IV response	QP density, parity switching, relaxation rate, offset charge, qubit-frequency shifts

The correct reduced full-project chain becomes

$$\begin{array}{c}
 \text{Module 1: radiation energy deposition} \\
 \Downarrow \\
 \begin{array}{cc}
 \text{Module 3: traps, TLSs, material disorder} & \text{Module 4: phonons and QP generation} \\
 \Downarrow & \Downarrow \\
 \text{Module 2: electrostatic weighting and offset-charge coupling} \\
 \Downarrow \\
 \text{Module 5: } Q_{\text{off}}(t), n_{\text{qp}}(\mathbf{x}, t), \Gamma_1^{\text{qp}}(t), \Gamma_{\text{parity}}(t), \delta\omega_q(t)
 \end{array}
 \end{array} \tag{20.1}$$

21 Summary of modeling replacements

The following replacements should guide future code development:

Baseline Module 5 object		Superconducting Module 5 replacement
Room-temperature mobile n, p	→	Cryogenic transient substrate charge $n_{\text{sub}}, p_{\text{sub}}$ plus trap occupations.
SRH trap recombination only	→	Trap occupation, offset charge, and TLS switching.
Semiconductor drift current	→	Supercurrent plus QP tunneling and poisoning response.
Temperature-dependent mobility only	→	QP generation, recombination, trapping, and gap/disorder changes.
Leakage-current output	→	QP density, parity switching, T_1 degradation, resonator response, and Q_{off} .

Bottom line. Module 5 in the superconducting region should be a hybrid charge-response module. Keep ordinary drift-diffusion only for transient radiation-generated charges in non-superconducting regions. Add quasiparticle diffusion-reaction in superconducting films. Add Josephson-junction response maps from junction-local QP density to poisoning and relaxation. Add offset-charge maps from trapped charge to island charge using Module 2 weighting potentials. This gives RadiationEffects_proj2 a physically meaningful bridge from radiation transport to superconducting-device degradation.

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